

$$\sum_{n=2}^{\infty} \log\left(1 + \frac{(-1)^n}{\sqrt{n}}\right)$$

$\Rightarrow \varepsilon > 0$. 如果 $|x| \leq \varepsilon$

$$\left| \log(1+x) - \left(x - \frac{x^2}{2}\right) \right| \leq |x|^3$$

$\exists N \in \mathbb{N}$ 如果 $n \geq N$

$$\left| \underbrace{\log\left(1 + \frac{(-1)^n}{\sqrt{n}}\right)}_{\uparrow \text{发散}} - \left(\underbrace{\frac{(-1)^n}{\sqrt{n}}}_{\uparrow \text{收敛}} - \underbrace{\frac{1}{2n}}_{\uparrow \text{发散}} \right) \right| \leq \underbrace{\frac{1}{n^{\frac{3}{2}}}}_{\uparrow \text{收敛}}$$

$$1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1} > \frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2k} \right) \quad \text{级数发散}$$

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

$$\neq \left(1 - \frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{6} - \frac{1}{8} \right) + \left(\frac{1}{5} - \frac{1}{10} - \frac{1}{12} \right) + \dots$$

$$= \left(\frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{6} - \frac{1}{8} \right) + \left(\frac{1}{10} - \frac{1}{12} \right) + \dots$$

$$= \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right) = \frac{1}{2} \log 2$$

要求: 绝对收敛级数

柯西乘积(幂级数乘法)

两个绝对收敛的幂级数 $\sum_{n=0}^{\infty} a_n x^n$, $\sum_{n=0}^{\infty} b_n x^n$ 乘积

$$\left(\sum_{n=0}^{\infty} a_n x^n\right) \left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k \cdot b_{n-k}\right) \cdot x^n$$

对于 $\left(\sum_{n=0}^{\infty} x^n\right)^2$ ($|x| < 1$)

$$\sum_{n=0}^{\infty} \sum_{k=0}^n x^k \cdot x^{n-k} = \sum_{n=0}^{\infty} (n+1) x^n = \left(\frac{1}{1-x}\right)^2 = \left(\sum_{n=0}^{\infty} x^n\right)^2$$

$$\sum_{n=1}^{\infty} \frac{\sin nx}{n} = \begin{cases} 0 & x=0 \\ \frac{\pi-x}{2} & x \in [0, \pi] \end{cases}$$

取 $a_n = \frac{1}{n} \downarrow 0$

$$b_n = \sin nx \quad B_n = \sum_{k=1}^n \sin kx$$

$$B_n \cdot 2 \sin \frac{x}{2} = \sum_{k=1}^n \left| \cos \left(k - \frac{1}{2}\right)x - \cos \left(k + \frac{1}{2}\right)x \right|$$

$$\log(1-z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \dots$$

取 $z = e^{i\theta}$

$$\text{则 } \log \underset{\substack{\uparrow \\ a-bi}}{1-e^{i\theta}} = -\sum_{n=1}^{\infty} \frac{1}{n} e^{in\theta}$$

$$1 - e^{i\theta} = e^{a-bi} = e^a (\cos b - i \sin b)$$

$$\begin{cases} 1 - \cos \theta = e^a \cos b \\ \sin \theta = e^a \sin b \end{cases}$$

$$\tan b = \frac{\sin \theta}{1 - \cos \theta} \Rightarrow b = \frac{\pi - \theta}{2}$$

$$2 - 2 \cos \theta = e^{2a} \Rightarrow a = \log 2 \sin \frac{\theta}{2}$$

$$\sum_{n=1}^{\infty} \frac{\cos n\theta}{n} = -\log 2 \sin \frac{\theta}{2} \quad 0 < \theta < \pi$$

$$\sum_{n=1}^{\infty} \frac{\sin n\theta}{n} = \frac{\pi - \theta}{2} \quad 0 < \theta < \pi$$

$\sum_{n=1}^{\infty} \frac{\sin n}{n}$ 条件收敛

① 证明收敛性

狄利克雷判别法：若数列 $\{a_n\}$ 单调递减且 $\lim_{n \rightarrow \infty} a_n = 0$ ，且级数 $\sum_{n=1}^N b_n$ 部分和有界，则 $\sum_{n=1}^{\infty} a_n b_n$ 收敛

令 $a_n = \frac{1}{n}$ 单调递减且 $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

$$b_n = \sin n \quad \text{部分和为} \quad S_N = \sum_{n=1}^N \sin n = \sum_{n=1}^N \frac{\cos(n - \frac{1}{2}) - \cos(n + \frac{1}{2})}{2 \sin \frac{1}{2}} = \frac{\cos \frac{1}{2} - \cos(N + \frac{1}{2})}{2 \sin \frac{1}{2}}$$

$$\Rightarrow |S_N| < \frac{1}{|\sin \frac{1}{2}|} \quad \text{部分和有界}$$

故 $\sum_{n=1}^{\infty} \frac{\sin n}{n}$ 收敛

② 证明不绝对收敛

$$\sum_{n=1}^{\infty} \left| \frac{\sin n}{n} \right| \text{ 收敛}$$

$$\text{则有 } \left| \frac{\sin n}{n} \right| \geq \frac{\sin^2 n}{n} = \frac{1}{2n} - \frac{\cos 2n}{2n}$$

$$\sum_{n=1}^{\infty} \frac{1}{2n} \text{ 是发散的}$$

而由狄利克雷判别法, $\sum_{n=1}^{\infty} \frac{\cos 2n}{2n}$ 收敛

$$\text{故 } \sum_{n=1}^{\infty} \left| \frac{\sin n}{n} \right| \text{ 发散}$$

故 $\frac{\sin n}{n}$ 条件收敛

证明级数 $\sum_{n=1}^{\infty} \frac{\cos n}{n}$ 收敛, 而 $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n}$ 发散

$\{\frac{1}{n}\}$ 单调递减且趋于零

$$\begin{aligned} \left| \sum_{k=1}^n \cos k \right| &= \frac{1}{2 \sin \frac{1}{2}} \left| \sum_{k=1}^n 2 \cos k \cdot \sin \frac{1}{2} \right| \\ &= \frac{1}{2 \sin \frac{1}{2}} \left| \sum_{k=1}^n \left(\sin \frac{2k+1}{2} - \sin \frac{2k-1}{2} \right) \right| \\ &= \frac{1}{2 \sin \frac{1}{2}} \left| \sin \frac{2n+1}{2} - \sin \frac{1}{2} \right| \leq \frac{1}{\sin \frac{1}{2}} \end{aligned}$$

有界, 故 $\sum_{n=1}^{\infty} \frac{\cos n}{n}$ 收敛

$$\text{而 } \sum_{n=1}^{\infty} \frac{\cos^2 n}{n} = \sum_{n=1}^{\infty} \left(\frac{1}{2n} + \frac{\cos 2n}{2n} \right) \quad \text{中 } \sum_{n=1}^{\infty} \frac{1}{2n} \text{ 发散, } \sum_{n=1}^{\infty} \frac{\cos 2n}{2n} \text{ 收敛}$$

故 $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n}$ 发散

